

★ DATA + CHARACTERISTICS + IPO ★

Computers | Ch01 Fundamentals + Devices | Topic 1 of 12

► Meaning in one line

Raw-data , processed-info , accuracy/speed/diligence/versatility/memory , Input-Process-Output.

► Key points

Point	Example
data-raw + info-processed; traits: accuracy/speed/diligence.	-
IPO cycle.	keyboard→CPU→monitor.

★ IMPORTANT FOR EXAM

Analog/digital/hybrid + micro-to-super

⚠ TRAP - read twice

Info-as-data + 'fast = smart' + IPO-miss

✓ **Try yourself:** Raw vs processed + IPO eg? **Answer:** Data/info + keyboard-CPU-monitor.

★ COMPUTER TYPES ★

Computers | Ch01 Fundamentals + Devices | Topic 2 of 12

► Meaning in one line

Analog , digital , hybrid , micro/mini/mainframe/super.

► Key points

Point	Example
analog/digital/hybrid.	-
micro/mini/main/super + PARAM.	-

★ IMPORTANT FOR EXAM

Generations 1-5

⚠ TRAP - read twice

Super-everywhere + 'mainframe dead' + size-mix

✓ **Try yourself:** Hybrid + supercomputer eg? **Answer:** Both + PARAM/weather.

★ GENERATIONS 1-5 ★

Computers | Ch01 Fundamentals + Devices | Topic 3 of 12

► Meaning in one line

Vacuum , transistor , IC , VLSI , AI.

► Key points

Point	Example
G1-vacuum + G2-transistor.	-
G3-IC + G4-VLSI + G5-AI.	-

★ IMPORTANT FOR EXAM

Bits/bytes + binary + ASCII

⚠ TRAP - read twice

G4-AI + 'generation = year' + tech-mix

✓ **Try yourself:** 3rd + 4th gen tech? **Answer:** IC; VLSI/microprocessor.

★ BITS + BINARY/ASCII ★

Computers | Ch01 Fundamentals + Devices | Topic 4 of 12

► Meaning in one line

Bit/nibble/byte , 0/1 , ASCII/Unicode.

► Key points

Point	Example
4/8-bits + KB/MB/GB.	1 byte = 8 bits.
binary + ASCII/Unicode.	A = 65.

★ IMPORTANT FOR EXAM

CPU/ALU/CU

⚠ TRAP - read twice

Byte-bit-mix + 'ASCII = binary'

✓ **Try yourself:** Nibble + 1KB bytes? A in ASCII? **Answer:** 4 bits; 1024; 65.

★ CPU + ALU + CU ★

Computers | Ch01 Fundamentals + Devices | Topic 5 of 12

► Meaning in one line

Brain , math , control , fetch/decode/execute.

► Key points

Point	Example
CPU/ALU/CU + fetch/decode/execute.	-
Registers/clock + speed.	-

★ IMPORTANT FOR EXAM

Registers + cache + motherboard

⚠ TRAP - read twice

ALU/CU-swap + 'CPU = box' + cycle-miss

✓ **Try yourself:** ALU vs CU + cycle order? **Answer:** Math vs control; fetch-decode-execute.

★ REGISTERS + CACHE + MOTHERBOARD ★

Computers | Ch01 Fundamentals + Devices | Topic 6 of 12

► Meaning in one line

Fastest , SRAM , board , slots , buses.

► Key points

Point	Example
Registers + fastest.	PC/MAR/MDR.
Cache/board + SRAM + slots.	-

★ IMPORTANT FOR EXAM

Buses + clock + keyboard

⚠ TRAP - read twice

RAM-cache-mix + 'motherboard = CPU'

✓ **Try yourself:** Fastest memory? Board holds? **Answer:** Registers; CPU/RAM/slots.

★ BUSES + CLOCK + KEYBOARD ★

Computers | Ch01 Fundamentals + Devices | Topic 7 of 12

► Meaning in one line

Lanes , GHz/cores , keys , shortcuts.

► Key points

Point	Example
Buses + clock/cores.	GHz + quad.
Keyboard + layout + shortcuts.	-

★ IMPORTANT FOR EXAM

Mouse + scanner + OCR/OMR/MICR

⚠ TRAP - read twice

Bus-library + 'cores = speed only' + QWERTY-miss

✓ **Try yourself:** Bus + core + QWERTY? **Answer:** Lanes; speed; layout.

★ MOUSE + SCANNER ★

Computers | Ch01 Fundamentals + Devices | Topic 8 of 12

► Meaning in one line

Point/click/scroll , photo/scan.

► Key points

Point	Example
Mouse/pointing.	optical/trackball/touchpad.
Scanner + flatbed.	photo-to-digital.

★ IMPORTANT FOR EXAM

OCR/OMR/MICR + barcode

⚠ TRAP - read twice

Scanner-printer-mix + 'mouse = CPU'

✓ **Try yourself:** Point + photo devices? **Answer:** Mouse; scanner.

★ OCR + OMR + MICR + BARCODE ★

Computers | Ch01 Fundamentals + Devices | Topic 9 of 12

► Meaning in one line

Text/marks/cheques , read.

► Key points

Point	Example
OCR/OMR/MICR.	text/bubbles/cheques.
Barcode/QR.	lines/squares + scan.

★ IMPORTANT FOR EXAM

Biometric + touch + monitor

⚠ TRAP - read twice

OCR-OMR-mix + 'barcode = QR' + use-miss

✓ **Try yourself:** Cheques + exams + text? **Answer:** MICR + OMR + OCR.

★ BIOMETRIC + TOUCH + MONITOR ★

Computers | Ch01 Fundamentals + Devices | Topic 10 of 12

► Meaning in one line

Body , feel , see.

► Key points

Point	Example
Biometric + body.	finger/face/iris.
Touch/monitor + feel/see.	tap + LCD/LED.

★ IMPORTANT FOR EXAM

Printers (impact/non)

⚠ TRAP - read twice

Password-only + 'touch = input-only' + LCD/LED-mix

✓ **Try yourself:** Body + feel + see devices? **Answer:** Biometric; touch; monitor.

★ PRINTERS: BASICS + IMPACT/NON-IMPACT ★

Computers | Ch01 Fundamentals + Devices | Topic 11 of 12

► Meaning in one line

Bang , spray.

► Key points

Point	Example
Impact + dot/daisy/line.	noisy + forms.
Non-impact + ink/laser.	quiet + photo.

★ IMPORTANT FOR EXAM

Till end: plotter + speakers + projector + compendium!

⚠ TRAP - read twice

Laser-impact + 'all same' + use-miss

✓ **Try yourself:** Forms + photo + types? **Answer:** Impact; non-impact.

★ PLOTTER + SPEAKERS + PROJECTOR ★

Computers | Ch01 Fundamentals + Devices | Topic 12 of 12

► Meaning in one line

Draw , sound , show.

► Key points

Point	Example
Plotter + vector.	maps/designs.
Speakers/projector.	sound/show.

★ IMPORTANT FOR EXAM

Memory hierarchy + RAM

⚠ TRAP - read twice

Printer-plotter-mix + 'speakers input' + lumens-miss

✓ **Try yourself:** Maps + sound + hall-show? **Answer:** Plotter; speakers; projector.